

Subtype-specific health and economic impact of delayed breast cancer diagnosis during the early COVID-19 pandemic in Belgium: a Markov model analysis

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Title: Subtype-specific health and economic impact of delayed breast cancer diagnosis during the early COVID-19 pandemic in Belgium: A Markov model analysis

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ABSTRACT (338 words)

Background: During the first COVID-19 wave, breast cancer diagnoses declined sharply worldwide due to suspended screening programmes and delayed care-seeking driven by fear of infection. In Belgium, organized screening programmes were halted from March to June 2020, leaving 135 invasive breast cancers undiagnosed. Although no stage shifts were observed in 2020, these undiagnosed cases risk later detection at more advanced stages, with worse prognosis, higher healthcare costs, and reduced health-related quality of life. Evidence indicates that such delays disproportionately affect aggressive subtypes (e.g., triple-negative (TNBC)) compared with slower-growing luminal-like cancers. This study projected the five-year impact of diagnostic delays on health outcomes and costs, stratified by molecular subtype.

Methods: A Markov cohort model compared two cohorts of 10,147 Belgian women with breast cancer: a “disrupted-care” cohort (2020 data, including 135 undiagnosed cases) and a “non-disrupted” cohort (2017–2019 trends). Outcomes over five years were estimated from the healthcare payer perspective, including incremental QALYs, direct medical costs, and mortality. Data sources included the Belgian Cancer Registry, literature, and national cost databases. Sensitivity and scenario analyses assessed uncertainty.

Results: Over five years, diagnostic delays were projected to cause a deterministic total loss of 21 QALYs and €3.2M in additional healthcare costs across all subtypes, resulting in six additional deaths. This corresponded to an average loss of 0.002 QALYs and €315 additional costs per patient. The burden was disproportionately carried by aggressive subtypes. TNBC accounted for the largest health loss (-9.5 QALYs) and highest incremental costs (€1.6M), followed by HER2+ cancer (-2.5 QALYs; €0.5M).

Probabilistic sensitivity analysis revealed considerable uncertainty in these estimates, particularly influenced by assumed input parameters.

Conclusion: The impact of diagnostic delays during Belgium's first COVID-19 wave was less severe than expected, likely reflecting rapid recovery measures that limited sustained stage shifts. While overall effects were modest, they may conceal a higher burden among faster-progressing subtypes such as TNBC and HER2+. Substantial model uncertainty highlights the need for routine registration of molecular subtypes to support future research on more tailored screening and diagnostic approaches during health system disruptions.

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1 | BACKGROUND

Cancer remains a leading cause of morbidity and mortality worldwide, placing a substantial health and economic burden on societies. In Europe alone, total cancer-related costs were estimated at €199 billion in 2018(1–3), including direct medical expenses, indirect costs (e.g., productivity losses), and direct non-medical costs like informal caregiving. Breast cancer is the most prevalent cancer among women and the second costliest cancer globally(4,5). Belgium has among the highest age-standardized breast cancer incidence rates in the world(6), contributing substantially to the national healthcare burden. In 2018, breast cancer healthcare costs exceeded €300 million(4). This is driven by high incidence—11,636 new cases in 2023(7)—and an aging population, which increases disease risk and care complexity(4). Despite the high costs, breast cancer has a relatively low burden of years lived with disability per case(4). This is largely due to early detection—often through organized screening programmes—and effective treatment, which improve prognosis and reduce recurrence and long-term disability(4). These outcomes are further supported by Belgium’s structured breast cancer care pathway, with accredited clinics, mandatory multidisciplinary consultations, and national standards for timely, coordinated, high-quality care(7).

During the first wave of COVID-19, many countries implemented strict public health measures, including lockdowns, social distancing, and healthcare resource reallocation, to contain the spread of the virus(8). These led to suspension of non-urgent services and disrupted breast cancer care globally, with halted screening and adapted treatments to reduce exposure and preserve capacity(9). There was also concern that patients delayed seeking care due to fear of infection, healthcare strain, or a desire not to burden the system(10,11). Delayed breast cancer care could theoretically lead to more advanced-stage

presentations, resulting in higher treatment costs, reduced health-related quality of life (HRQoL), and poorer survival(12,13).

In Belgium, the organized breast cancer screening programme was suspended from March to June 2020, causing a 44% drop in Ductal Carcinoma In Situ (DCIS) diagnoses(14). As DCIS is mainly detected through screening, this decline reflects programme disruption, though some were still found symptomatically. By year-end, invasive breast cancer diagnoses remained 5% below expected levels, despite screening resumption(14). Importantly, Belgian data show that time to treatment was largely maintained or even accelerated, with no increase in tumour size or nodal involvement among symptomatic cases(14).

Breast cancer is a heterogenous disease with multiple subtypes that differ in biological behaviour and clinical outcomes. The most widely used classification system relies on immunohistochemical profiling of estrogen, progesterone, and human epidermal growth factor receptors 2 (HER2). Based on these markers, breast cancers are commonly grouped into four subtypes (Table 1): luminal A, luminal B, HER2-positive (HER2+), and triple-negative (TNBC). This classification guides treatment decisions and reflects differences in prognosis and response to therapy(15,16). Luminal A cancers tend to grow slowly and have a favourable prognosis, whereas HER2+ and TNBC subtypes are more aggressive, with faster progression and worse outcomes if diagnosis is delayed(11,17). In Belgium, data from 2014 show that luminal A-like tumours accounted for 54% of cases, followed by luminal B-like (27%), TNBC (13%), and HER2+ (6%) breast cancers(18).

The impact of a temporary disruption of cancer screening on HRQoL, healthcare costs, and mortality across breast cancer molecular subtypes in Belgium remains unknown. However, evidence from other countries indicates that even short-term screening interruptions may result in more advanced-stage diagnoses(19) and higher projected mortality(20–23), particularly among aggressive subtypes. These delays can also lead to substantial long-term HRQoL and economic burdens(23,24).

A recent study(25) found that even short diagnostic delays can lead to more advanced cancer, especially for aggressive subtypes. For TNBC, a delay of 38 days might lead to over 10% more patients being diagnosed at a later stage. This window was 52 days for HER2+ cancers but a longer 85 days for slower-growing luminal-like cancers, highlighting the heightened sensitivity of aggressive tumours to diagnostic delays.

Although Belgian data did not indicate tumour or nodal progression among breast cancer cases with delayed diagnosis between March and June 2020, these analyses were not stratified by molecular subtype(14). As such, potential stage progression in aggressive subtypes, such as TNBC or HER2+ cancers, may have been masked by the predominance of slower-growing tumours like luminal A.

Through a Markov cohort simulation model, we explore the potential hidden impact of these diagnostic delays on changes in Quality-Adjusted Life Years (QALYs) and healthcare costs, across molecular subtype, compared to pre-COVID-19 care conditions. In line with Belgian guidelines for health economic evaluations(26), these standard metrics were selected to capture the comprehensive patient health burden and the direct economic impact on the healthcare system, respectively. We projected the impact over a five-year horizon, capturing the period of highest recurrence risk for aggressive subtypes such as TNBC and HER2+(27). This timeframe also aligns with the most robust clinical evidence on outcomes and costs(15,16). In addition, a 10-year horizon was examined in scenario analysis to account for the more prolonged recurrence risk observed in luminal subtypes(27). These findings aim to provide healthcare professionals and policymakers with evidence to safeguard continuity of diagnostic services during future system shocks, minimise backlogs once services resume, prioritise timely detection of aggressive

subtypes such as TNBC and HER2+ cancers, and encourage further breast cancer research stratified by molecular subtype.

2 | METHODS

We used a Markov cohort simulation model approach, as described in our protocol (28), to project over five years the health and healthcare cost impacts of COVID-19-attributable breast cancer diagnostic disruptions that occurred from mid-March to June 2020 in Belgium. Markov models effectively capture breast cancer's chronic nature, stage progression, and long-term recurrence and mortality risk(29). The analysis was stratified by four breast cancer molecular subtypes (Table 1): luminal A, luminal B, HER2+, and TNBC.

Table 1. Characteristics of the four breast cancer molecular subtypes (17)

Molecular subtype	Receptor status	Biological behaviour	Prognosis	Treatment
Luminal A	ER+, PR+, HER2–, low Ki-67	Slow-growing, hormone-sensitive	Favourable	Endocrine therapy
Luminal B	ER+, PR±, HER2– or HER2+, high Ki-67	More proliferative than Luminal A	Intermediate	Endocrine therapy + CDK4/6 inhibitors, ± chemotherapy
HER2+	HER2+, ER–/low, PR–/low	Highly proliferative, treatment-sensitive	Variable (improved with HER2-targeted therapy)	Anti-HER2 therapy (e.g., trastuzumab, pertuzumab) + chemotherapy
TNBC	ER–, PR–, HER2–	Highly aggressive, rapid progression	Poor	Chemotherapy ± immunotherapy

ER: Estrogen receptor; PR: Progesterone receptor; HER2: Human Epidermal Growth Factor Receptor 2; TNBC: Triple-negative breast cancer.

**Ki-67 = a marker of cellular proliferation, used to distinguish Luminal A (low Ki-67) from Luminal B (high Ki-67)*

***CDK4/6 = cyclin-dependent kinases 4 and 6, targeted by specific inhibitors in HR+/HER2– breast cancer to block cell cycle progression.*

2.1 | Study design

Aggregated, de-identified incidence data by clinical stage were obtained from the Belgian Cancer Registry (BCR, 2024), which compiles nationwide data from oncology programmes and pathology laboratories (age, stage, tumour and patient characteristics)(30). Ethical Committee approval was not required by law, since the collection of the data and making them available was performed within the legal framework of BCR (legal basis: art. 138 Coordinated Law of 10 May 2015), and since this retrospective study does not fall under the Belgian Law of 7 May 2004 regarding experiments on human persons (art. 3,§2) which only requires EC approval for prospective studies.

Subtype-specific Markov cohort models were developed, simulating the movement of two closed incident cohorts of 10,147 adult female breast cancer patients over five years. These included a non-disrupted care cohort, estimated from 2017–2019 trends using a Poisson model, and a disrupted-care cohort reflecting observed 2020 cases. Predicted stage-specific values were adjusted to match the observed decline, ensuring equal cohort sizes for unbiased comparison. The comparison between projected and observed data revealed a shortfall of 218 cases in Stage I, which was partially offset by slight observed excesses in Stages II–IV (see Table 2). The core of the diagnostic disruption was represented by the Undiagnosed Cancer (UDC) cohort, defined as the net deficit of 135 cases derived by subtracting the total observed stage I–IV cases in 2020 (10,012) from the total predicted incidence (10,147).

COVID-19–related diagnostic disruption was represented by 135 undiagnosed cancer (UDC) patients that should have been diagnosed in 2020 based on 2017–2019 predictions. Two subtype-specific Markov cohort models were developed, each simulating over five years the movement of two closed incident cohorts of 10,147 adult female breast cancer patients: a non-disrupted care cohort, estimated from 2017–2019 trends using a Poisson model, and a disrupted-care cohort reflecting observed 2020 cases (symptomatic, opportunistic, and resumed screening). Predicted stage-specific values were adjusted to match the observed decline, ensuring equal cohort sizes for unbiased comparison. UDC were derived by subtracting observed stage I–IV cases in 2020 from the corrected predictions.

Age stratification was not applied, as molecular subtype is the primary driver of prognosis and treatment decisions(15,16). While age may influence care through comorbidities or patient preference, including it would have added complexity without improving model validity. Because subtype-specific incidence data were unavailable, pre-COVID Belgian subtype proportions were applied to both cohorts by multiplying each stage count by the corresponding subtype share to estimate per-subtype counts.

A five-year time horizon was chosen to capture the period of highest recurrence risk and cost concentration, particularly for aggressive subtypes such as TNBC and HER2+(27), and to align with follow-up guidelines and available evidence. A three-month cycle length was used, reflecting typical treatment intervals (e.g., chemotherapy cycles, clinical follow-ups) (15). The model adopted a Belgian healthcare payer perspective, focusing on direct medical costs (e.g., treatments, consultations). This approach aligns with the National Institute for Health and Disability Insurance (NIHDI) reimbursement practice and avoids the added uncertainty that would arise from applying a societal perspective, given the lack of subtype-specific data on productivity losses and informal caregiving.

The model was developed in Excel, with patients distributed at baseline (cycle 0) across the initial health states—‘Stage I Diagnosis & Treatment (D&T)’, ‘Stage II D&T’, ‘Stage III D&T’, ‘Stage IV D&T’, and ‘UDC’

(see Figure 1)—according to BCR incidence data. Each cohort then progressed through health states via a transition matrix, which applied predefined probabilities to redistribute patients every three months. Each health state was associated with specific healthcare costs and utilities (see 2.4 Data inputs). By iterating these cycles, the simulation generated a dynamic representation of the patient population's trajectory over the five-year horizon. Ultimately, the model compared QALYs (where 1 = perfect health and 0 = death) and healthcare costs between cohorts. Input parameters for the simulation, including transition probabilities, healthcare costs, and utilities, are detailed in Table 4 (presented later in the manuscript), with calculation methods provided in the Additional file 1.

Table 2. Breast cancer cases by stage and subtype in non-disrupted (2017–2019) and disrupted (2020) cohorts (aggregated data from the Belgian Cancer Registry, 2024)

Non-disrupted care cohort (Pre-COVID-19 situation, 2017-2019)					
Stage (clinical) at diagnosis	Total	Luminal A	Luminal B	HER2-positive	Triple-negative
Stage I	4 769 (47.00%)	2 575	1 288	286	620
Stage II	3 904 (38.47%)	2 108	1 054	234	508
Stage III	716 (7.06%)	387	193	43	93
Stage IV	758 (7.47%)	409	205	45	99
Total	10 147	5 479	2 740	609	1 319
Disrupted-care cohort (COVID-19 situation, 2020)					
Stage (clinical) at diagnosis	Total	Luminal A	Luminal B	HER2-positive	Triple-negative
Stage I	4 551 (44.85%)	2 458	1 229	273	592
Stage II	3 949 (38.92%)	2 132	1 066	237	513
Stage III	753 (7.42%)	407	203	45	98
Stage IV	759 (7.48%)	410	205	46	99
Undiagnosed Cancer	135 (1.33%)	73	36	8	18
Total	10 147	5 479	2 740	609	1 319

Note: Values represent aggregated patient counts and have been rounded; minor discrepancies may occur. Subtype-specific counts were estimated by applying pre-COVID Belgian subtype proportions(18) to stage counts. In the absence of subtype-specific stage incidence data, we assumed that each subtype followed the overall stage distribution of breast cancer.

2.2 | Markov Model structure

The model consisted of 11 health states (Figure 1), grouped into non-metastatic (stages I–III) and metastatic (stage IV) disease. Non-metastatic disease begins with a 'diagnosis and treatment' (D&T) phase, which includes diagnostic work-up, neoadjuvant therapy (for HER2+ and TNBC), surgery (with reconstruction if applicable), and radiotherapy. This phase is represented in the model as 'Stage I D&T,' 'Stage II D&T,' or 'Stage III D&T,' depending on the initial stage at diagnosis. Patients remain in the D&T state for one or more cycles depending on their subtype-specific treatment duration (Table 3), before transitioning to 'Stage I–III Follow-Up (FU)' for a defined period of adjuvant therapy (AT). We modelled AT as five years (20 cycles) for luminal A and seven years (28 cycles) for luminal B, reflecting standard practice for hormone receptor–positive disease and extended therapy increasingly recommended for luminal B due to its higher risk of late recurrence(31).

Given the differing nature and duration of AT across molecular subtypes, a distinct 'Stage I–III FU' health state was defined. After AT, patients enter this state with follow-up every three months until recurrence. Some, however, may not achieve a disease-free period and directly experience disease progression. In both cases, we simplify this by saying that patients move to 'Stage I–III Locoregional Recurrence (LRR)' if the evolution is locoregional or to 'Stage IV D&T' if metastatic. Those in 'Stage I–III LRR' receive treatment during one cycle and then return to follow-up, while patients entering the metastatic pathway

(Stage IV states) start systemic therapy in ‘Stage IV D&T’ and transition through subsequent lines of treatment as disease advances.

Table 3. Duration spent in initial treatment-related health states by molecular subtype, independent of COVID-19 disruptions.

Health States	Luminal A	Luminal B	HER2+	Triple-Negative
Stage I D & T	3 months (1 cycle)	3 months (1 cycle)	9 months (3 cycles)	9 months (3 cycles)
Stage II D & T	3 months (1 cycle)	3 months (1 cycle)	9 months (3 cycles)	9 months (3 cycles)
Stage III D & T	3 months (1 cycle)	3 months (1 cycle)	9 months (3 cycles)	9 months (3 cycles)
Stage I-III FU (Adjuvant therapy)	5 years (20 cycles)	7 years (28 cycles)	1 year (4 cycles)	1 year (4 cycles)

D&T: diagnosis and treatment; FU: follow-up

Note: Stage IV (metastatic) disease is not included in this table, as its clinical course typically involves successive lines of systemic therapy and differs from the more structured, stage-based pathways of earlier stages. While treatment-free or even disease-free intervals may occur, these patterns were not incorporated in the model for reasons of simplification. The durations listed here reflect initial, curative-intent treatment for non-metastatic disease (stages I–III) prior to progression.

Patients diagnosed de novo with metastatic disease also start at ‘Stage IV D&T’, where they receive a first line of systemic therapy. They remain in this state (illustrated through the looping arrow) until disease progression, where they directly transition to the second-line regimen. This process continues sequentially up to four lines of metastatic treatment: ‘Stage IV 2L’, ‘3L’, and ‘4L’. A separate health state,

'Undiagnosed Cancer' (UDC), captures patients whose diagnosis was delayed due to COVID-19-related disruptions and who can subsequently transition to any stage at diagnosis once detected. During any cycle, patients in any health state can transition to 'Dead' (absorbing state).

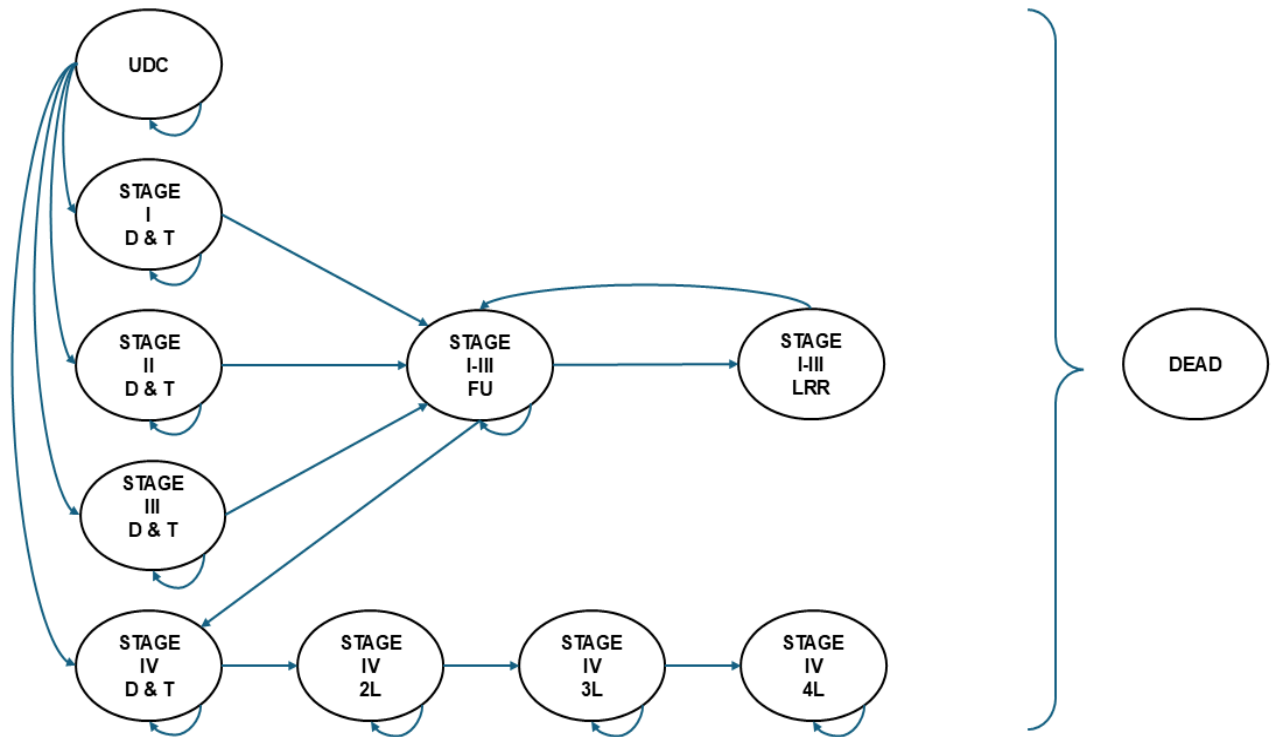


Figure 1. Breast cancer Markov model diagram with 11 health states, applicable to each molecular subtypes.

D & T: Diagnosis and treatment; FU: Follow-up; L: Lines of systemic therapy; LRR: Locoregional recurrence

L: Lines of systemic therapy; UDC : Undiagnosed cancer

$\curvearrowright$ = Probability of remaining in the state

$\rightarrow$ = Probability of moving to another state

$] =$ Probability for patients in any state to move to the dead state

2.3 | Model validation

Clinical validation was conducted through consultation rounds with oncologists C.M. and E.N., to validate the model structure (i.e., model diagram), input parameters (i.e., healthcare costs), and key assumptions

(i.e., simplified treatment pathways), ensuring alignment with current clinical practice and expert opinion. Additionally, five- and ten-year mortality outcomes from the non-disrupted care model were validated against published subtype-specific survival rates⁽³²⁾, converted to mortality values (mortality = 1 – survival), which were consistent with our estimates.

2.4 | Data inputs

To operationalise our Markov model, aggregated incidence data from the BCR were used, transition probabilities and utilities were extracted from the literature, and healthcare costs were sourced from the Belgian medical nomenclature database (NomenSoft) and the Belgian pharmacotherapeutic information database (BCPI). All input values were stratified by health state and molecular subtype. The selected input values are presented in Table 4, with calculation methods and rationale detailed in the Additional file 1.

Both models used the same input values, except for aggregated incidence data and transition probabilities from the UDC state to stage I–IV D&T. In the non-disrupted care cohort, there were no undiagnosed breast cancers related to care disruptions, and thus no transitions from the UDC state occurred. In contrast, the disrupted care model included UDC cases with hypothesized transition probabilities to stage I–IV D&T to reflect diagnostic delays.

Insert table 4 here

2.4.1 | Epidemiological data

As mentioned above, stage-specific incidence data for breast cancer diagnoses for the disrupted and non-disrupted care cohorts were obtained from the BCR, enabling the estimation of UDC (they represent stage-specific differences between cohorts). In the absence of subtype-specific stage incidence data, we

assumed that each subtype followed the overall stage distribution of breast cancer, applying pre-COVID Belgian subtype proportions(18) to derive per-subtype estimates. Under 5% of diagnoses in the BCR dataset lacked definitive staging (unknown stage). Reasons for incomplete staging are hypothesized to differ between non-disrupted care cohort (e.g., frail elderly) and disrupted care cohort (e.g., potentially, interrupted diagnostics due to intensive care unit admission). Given the uncertainty in distribution, the minimal overall percentage, and the absence of a statistically significant difference between cohorts, unknown stage cases were excluded from the main model to simplify comparisons.

2.4.2 | Transition probabilities

The 3-month subtype-specific transition probabilities were mainly derived from international retrospective cohort studies (Table 4). We hypothesized 3-month transition probabilities from UDC to 'Stage I–IV D&T' and 'Dead' for each molecular subtypes—given the absence of subtype-specific incidence data on UDC—based on a study which showed that aggressive subtypes (i.e., TNBC and HER2+) reach stage progression much earlier than luminal-like cancers (25).

2.4.5 | Healthcare costs

Healthcare costs for each health state were estimated from the perspective of the Belgian healthcare payer (NIHDI). The model's costs represent the official reimbursed amounts (nomenclature codes 1300/1600). Patient co-payments (codes 3300/3600) were excluded to reflect only the payer's expenditure. Diagnostic, surgical, reconstruction, and radiotherapy costs were retrieved from NomenSoft(33), which provides official reimbursement rates (codes are available in the Additional file 1). Subtype-specific drug costs for neoadjuvant and AT were obtained from the Belgian Centre for Pharmacotherapeutic Information (BCPI)(34), with dosages and schedules based on phase III trials and

validated by two oncologists (C.M., E.N.). All reimbursement values were indexed to 2020 prices using the Belgian health index (35) to ensure consistency with the model's reference year.

Subtype-specific treatment pathways (Table 5) were simplified to balance between typical clinical practice and ensure feasibility within the Markov framework, and were based on ESMO clinical practice guidelines(15,16), which underpin Belgian national recommendations. Some treatment regimens were generalized, and treatment durations were averaged across patient groups. A standardized diagnostic pathway was assumed across all molecular subtypes, including biopsy, staging imaging, blood analysis, and mammography. For simplicity, we assumed only patients with HER2+ and TNBC to have subtype-specific neoadjuvant therapy. Surgery was stratified by stage: All stage I-II patients underwent breast-conserving surgery (BCS) without reconstruction, while for stage III patients, 70% underwent BCS and 30% underwent mastectomy, with 25% assumed to receive reconstruction. For simplicity, surgery was assumed to remain unchanged during the pandemic. All patients received the standard hypofractionated whole-breast radiotherapy regimen of 15 fractions, and related consultations before, during, and after treatment. AT also differed based on the molecular subtype. Post-adjuvant therapy follow-up included annual mammography, ultrasonography, and four consultations per year. For stage I-III patients developing LRR, healthcare costs were assumed to mirror those of initial D&T, calculated as the mean cost across stages I-III. Metastatic disease treatments followed subtype-specific systemic therapy lines, with healthcare costs also estimated on a 3-month cycle basis for comparability across subtypes. For simplicity, luminal B was assumed to have the same costs as luminal A.

Table 5. Simplified treatment pathway across health states and molecular subtype

Health state	Luminal-like	HER2+	TNBC
Stage I D & T	Diagnosis + BCS + RT (+ pre, per, post consultations)	Diagnosis + Neoadjuvant therapy + BCS + RT (+ pre, per, post consultations)	Diagnosis + Neoadjuvant therapy + BCS

			+ RT (+ pre, per, post consultations)
Stage II D & T	Diagnosis + BCS + RT (+ pre, per, post consultations)	Diagnosis + Neoadjuvant therapy + BCS + RT (+ pre, per, post consultations)	Diagnosis + Neoadjuvant therapy + BCS + RT (+ pre, per, post consultations)
Stage III D & T	Diagnosis + 70% BCS and 30% MST (+ 25% reconstruction) + RT (including pre, per, post consultations)	Diagnosis + Neoadjuvant therapy + 70% BCS and 30% MST (+ 25% reconstruction) + RT (including pre, per, post consultations)	Diagnosis + Neoadjuvant therapy + 70% BCS and 30% MST (+ 25% reconstruction) + RT (including pre, per, post consultations)
Stage I-III FU (Adjuvant therapy)	Adjuvant endocrine therapy: Either: tamoxifen/letrozole/ Exemestane/ anastrozole + triptorelin/goserelin (if premenopausal women) + CDK4 inhibitors: Abemaciclib	Chemotherapy + Anti-HER2 therapy (+ Endocrine therapy if HER2+)	Immunotherapy and Chemotherapy
Stage I-III FU (Post-adjuvant therapy)	Mammography Ultrasonography 4 consultations	Mammography Ultrasonography 4 consultations	Mammography Ultrasonography 4 consultations
Stage I-III LRR	Same as 'Stage I-III D&T'	Same as 'Stage I-III D&T'	Same as 'Stage I-III D&T'
Stage IV D & T	Diagnosis + endocrine therapy and CDK4/6 inhibitors	Anti-HER2 and Chemotherapy	Immunotherapy and Chemotherapy
Stage IV 2L	Targeted therapy	Anti-HER2	Targeted therapy
Stage IV 3L	Chemotherapy for patients with endocrine resistance	Chemotherapy and Anti-HER2	Chemotherapy
Stage IV 4L	Chemotherapy for patients with endocrine resistance	Chemotherapy and Anti-HER2	Chemotherapy

BCS: breast-conserving surgery; CDK4/6 inhibitors: cyclin-dependent kinases 4 and 6; D&T: Diagnosis and treatment; FU: Follow-up; HER2: Human epidermal growth factor receptor 2–positive; LRR: Locoregional recurrence; MST: Mastectomy; RT: Radiotherapy

Note. The breast cancer Markov model Excel file (Supplementary material), includes a detailed sheet with the exact treatment cost calculations.

2.4.6 | Utilities

Health-state utilities were derived from EQ-5D-5L data(36–38) using a two-step approach. Baseline utilities for Belgian women of all ages were taken from the national Health Interview Survey (36). Health-state-specific disutilities from international studies (37,38) were then subtracted, assuming comparable disease impacts across similar countries (e.g., the Netherlands). When no published disutilities were available for specific health states or subtypes, values were estimated based on additional literature(39–55), considering subtype-specific treatment burden, disease progression, and HRQoL changes. Utilities for luminal B were assumed to be equal to those of luminal A. Utilities for the UDC reflected general population levels, as asymptomatic patients were undiagnosed and likely unaffected by disease-specific HRQoL.

2.6 | Analysis

By running both cohorts, incremental differences in QALYs and healthcare costs between the disrupted care and non-disrupted care situations were estimated across subtypes over a five-year time horizon. QALYs were estimated by multiplying the utility value of each health state by the number of patients in that state per cycle and the cycle length. Summing these values across all cycles yielded the total QALYs. Healthcare costs were calculated similarly, by applying cost estimates to each state and aggregating them over all cycles.

To reflect present-day valuations, both healthcare costs and QALYs were discounted annually according to Belgian guidelines—3.0% for costs and 1.5% for health effects (26). Additionally, the analyses were used to calculate the percentage point differences in mortality and the estimated number of additional deaths attributable to delayed diagnosis across molecular subtypes.

2.7 | Scenario and sensitivity analyses

A one-way sensitivity analysis (OSA) and a probabilistic sensitivity analysis (PSA) were performed to assess parameter uncertainty and evaluate the robustness of results across each molecular subtype. The OSA included transition probabilities, utilities, and direct medical costs. Each parameter was varied individually across a range defined by its 95% confidence interval, calculated as the base-case value $\pm 1.96 \times$ standard error (SE), while holding all other parameters constant. Two tornado diagrams were produced: one showing the impact of varying transition probabilities and utilities on incremental QALYs, and another showing the impact of varying transition probabilities and treatment costs on incremental healthcare costs. Both diagrams reflect results combined across all molecular subtypes.

For the PSA, beta distributions were applied to probabilities and utilities, and gamma distributions to costs. Distribution parameters were derived from base-case values and their SEs, calculated from reported sample sizes where available. For parameters without reported variance, plausible 95% ranges (for probabilities/utilities) or an assumed coefficient of variation of 0.10 (for costs) were used to reflect uncertainty. The PSA was conducted using Monte Carlo simulation with 1,000 iterations in Visual Basic for Applications (VBA) within Excel.

Three scenario analyses were conducted to assess structural and methodological assumptions. A 10-year time horizon was used to explore longer-term outcomes. This captures the early recurrence peaks in HER2+ and TNBC (around two years) as well as the slower recurrence pattern of luminal A and B (beyond 10 years)(27,56,57). The same utility and cost inputs per cycle were maintained, with adapted transition probabilities reflecting recurrence risks beyond five years (27). In the second scenario, we adjusted the base-case 3-month transition probabilities from the UDC state to “Stage I–IV D&T” for each molecular subtype. The base case assumed slower-growing subtypes (e.g., luminal A) were more often diagnosed at early stages, and aggressive subtypes (e.g., TNBC) at later stages, based on clinical evidence(12,25). This scenario applied subtype-specific multipliers (Table 4) to model more severe progression, increasing late-stage and reducing early-stage diagnoses. As HER2+ and TNBC already had high late-stage

probabilities, adjustments were minor, whereas larger multipliers were applied to luminal-like subtypes to test possible underestimation of progression severity. The third scenario examined the impact of changes in radiotherapy reimbursement tariffs during the COVID-19 pandemic. As part of emergency response strategies, several Belgian hospitals adopted hypofractionated radiotherapy (five fractions instead of 15) to reduce hospital visits, minimise infection risks, and optimise resource use(58). This protocol was reimbursed under a higher-cost category (€2,581.73 - nomenclature code: 444710-444721) rather than the standard category III (€1,999.67 in 2020 - nomenclature code: 444150-444161). Belgian data indicate that 8.2% of patients received five fractions in 2020(14). Exploring this scenario is relevant to capture the potential cost implications of shifts in clinical practice and reimbursement during the pandemic.

3 | Results

3.1 | Base case analysis

The Markov model estimated that COVID-19 related diagnostic delays in breast cancer across Belgium from March-June 2020, led to a total loss of 20.7 QALYs and €3.19 million in additional healthcare costs over five years, compared to a situation without disruption (Table 6). This corresponds to an average loss of €315 per patient and 0.002 QALYs per patient—representing less than one day of life in full health (far below the minimal clinically important difference (MCID) typically cited for oncology, around 0.03 QALYs per year per patient(59)). Deterministic model outputs are point estimates derived from fixed parameter values and do not yield confidence intervals; uncertainty in these estimates was instead explored through the PSA.

Among molecular subtypes, TNBC accounted for the largest QALY loss (-9.5), followed by luminal A (-4.8), luminal B (-3.9), and HER2+ (-2.5). TNBC patients also incurred the highest additional direct healthcare

costs (€1.64 million), followed by HER2+ (€547,000), luminal B (€524,689), and luminal A (€481,944). Per patient, the largest QALY loss was observed in TNBC (-0.007 QALYs, or 2.56 days of full health), followed by HER2+ (-0.004 QALYs, or 1.46 days of full health), and the luminal-like subtypes, both below one day of full health (-0.001 QALYs). TNBC patients incurred the highest additional per-patient costs (€1,243), followed by HER2+ (€898), luminal B (€191), and luminal A (€88).

Delayed cancer diagnoses resulted in a slight increase in mortality across all subtypes. The largest absolute increase in mortality occurred in TNBC (+0.20 percentage points), resulting in two additional deaths. Luminal A and B subtypes showed increases in mortality of 0.08 and 0.05 percentage points, corresponding to two and one additional death(s), respectively. HER2+ increased by 0.07 percentage points, also leading to one additional death. For TNBC, this corresponds to a 1.9% relative increase compared to non-disrupted care. Luminal A and B showed relative increases of 2.9% and 1.2%, respectively, while HER2+ increased by 0.9%. Relative changes in mortality were calculated as the percentage difference from the non-disrupted care situation.

Insert table 6 here

3.3 | One-Way Sensitivity Analysis (OSA)

This section presents the OSA results examining how variations in transition probabilities, costs, and utilities influence both changes in incremental QALYs and healthcare costs. Two tornado diagrams (Figures 2 and 3) are provided to illustrate the parameters driving the greatest changes.

3.3.1 | The impact of varying transition probabilities and utilities on incremental QALYs

The OSA on QALYs showed that the model was most sensitive to changes in the utilities for TNBC. Particularly, utilities for stage I D&T, stage I–III adjuvant therapy, stage IV D&T, and stage II D&T had the largest impact on incremental QALYs.

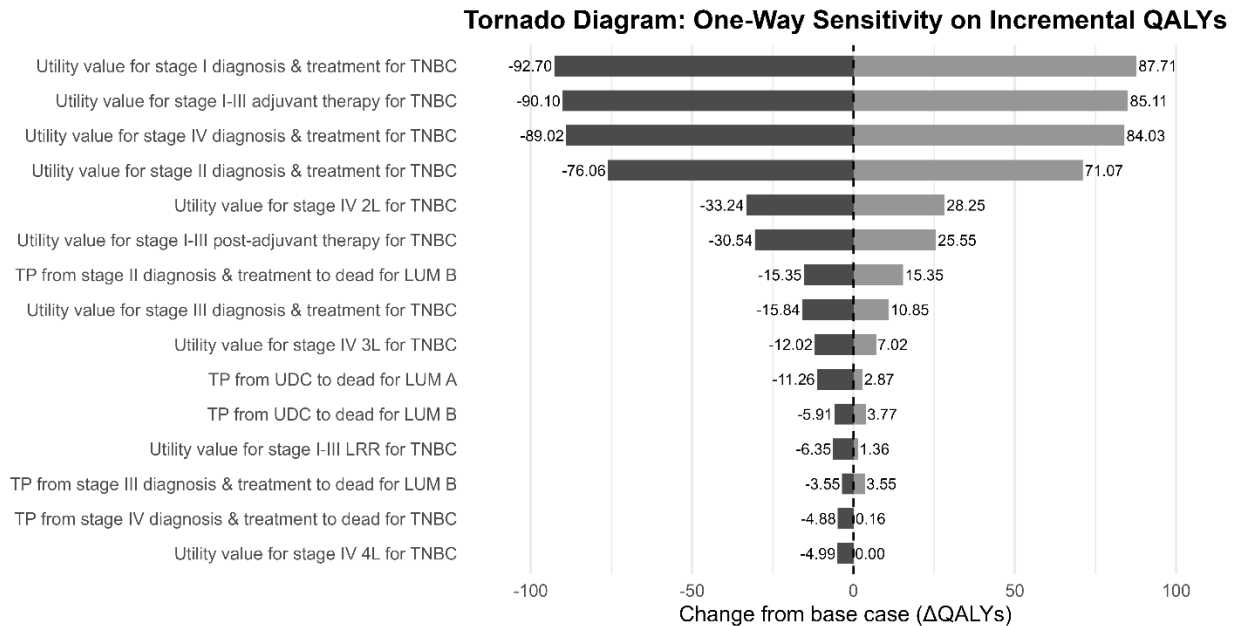


Figure 2. Tornado diagram showing the impact of varying transition probabilities and utilities on incremental QALYs across breast cancer molecular subtypes

L: Lines of systemic therapy (2L: second therapy line; 3L: third therapy line; 4L: fourth therapy line); LRR: Locoregional recurrence; LUM A: Luminal A; LUM B: Luminal B; QALY: Quality-Adjusted Life Year; TNBC: Triple-negative breast cancer; TP: Transition probabilities; UDC: Undiagnosed cancer

** The vertical dashed line represents no change from the base case ($\Delta QALY = 0$). Dark grey bars indicate the change when the parameter is set to its lower bound, and light grey bars indicate the change when set to its upper bound.*

3.3.2 | The impact of varying model inputs on incremental healthcare cost changes

The OSA on healthcare costs showed that the model was most sensitive to cost variations in TNBC.

Specifically, the costs of stage I D&T and stage IV D&T, which had the largest impacts on total incremental healthcare costs.

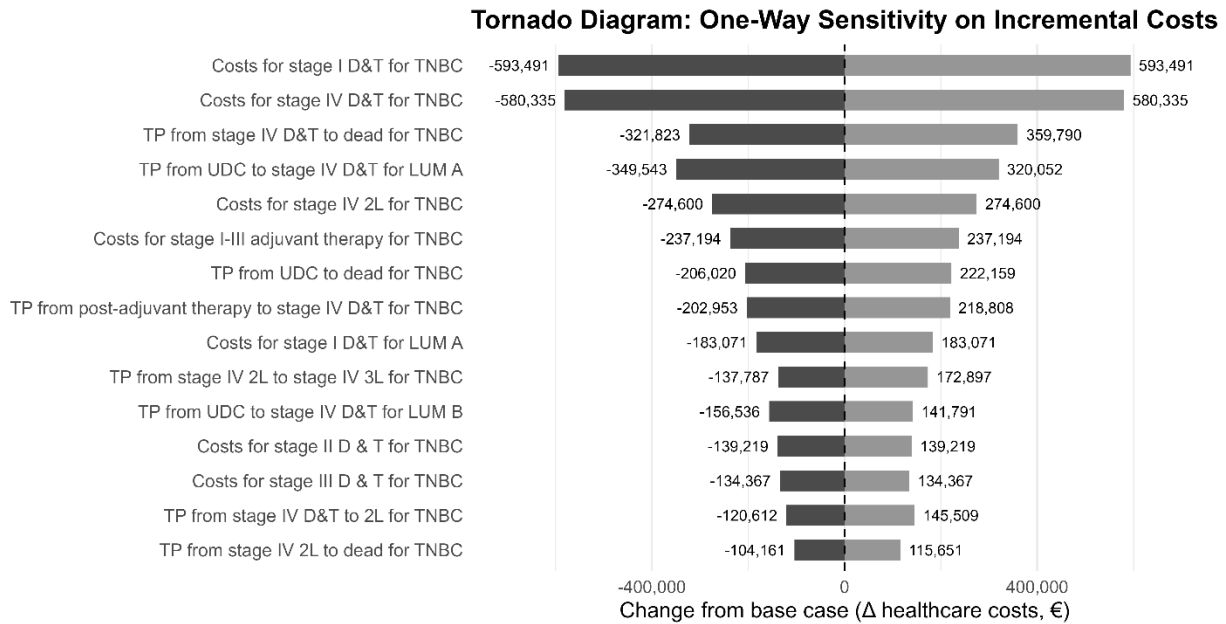


Figure 3. Tornado diagram showing the impact of varying transition probabilities and treatment costs on incremental healthcare costs across breast cancer molecular subtypes.

D&T: Diagnosis and treatment; L: Lines of systemic therapy (2L: second therapy line; 3L: third therapy line; 4L: fourth therapy line); LRR: Locoregional recurrence; LUM A: Luminal A; LUM B: Luminal B; TNBC: Triple-negative breast cancer; TP: Transition probabilities; UDC: Undiagnosed cancer

**The vertical dashed line represents no change from the base case (Δ healthcare costs = 0). Dark grey bars indicate the change when the parameter is set to its lower bound, and light grey bars indicate the change when set to its upper bound.*

3.4 | Scenario analyses

Extending the time horizon from five to 10 years revealed a more pronounced impact of COVID-19-related disruptions on both health and economic outcomes (Table 7). QALY losses widened from -20.7 to -32.1, and incremental healthcare costs rose by €2.50 million (from €3.19 million to €5.69 million). Luminal-like subtypes experienced over a twofold increase in both QALY losses and healthcare costs, whereas increases were less pronounced for TNBC and HER2+: QALY losses doubled for HER2+ but fell by more than half for TNBC, while costs rose by over 1.5-fold for both. Mortality trends were consistent with the base case, with TNBC showing the largest absolute increase (+0.33 percentage points), followed by HER2+ (+0.11), luminal B (+0.06), and luminal A (+0.04). These corresponded to

relative increases in mortality of 1.6% for TNBC (from 20.44% to 20.77%), 0.8% for HER2+ (from 13.64% to 13.75%), 0.7% for luminal B (from 9.10% to 9.16%), and 0.6% for luminal A (from 6.36% to 6.40%).

Applying increased transition probabilities through multipliers (Table 4) from the UDC state towards advanced stage at diagnosis (i.e., stage III and IV) across subtypes resulted in nearly 37 QALYs lost and €9 million in additional healthcare costs, almost a threefold increase compared to the base case (Table 6). Luminal A and B subtypes incurred €4.49 million and €1.82 million in costs, and lost 15.17 and 7.70 QALYs, respectively. TNBC costs increased by €371,181 (to €2.01 million), and HER2+ by €136,165 (to €683,165). This scenario projected 11 additional deaths—almost twice the number in the base case—mainly among luminal subtypes. Absolute mortality increases were highest in TNBC (+0.22 percentage points), followed by luminal A and HER2+ (+0.08 each) and luminal B (+0.09). These corresponded to relative mortality increases of 2.0% for TNBC (from 10.81% to 11.03%), 2.9% for luminal A (from 2.80% to 2.88%), 2.2% for HER2+ (from 8.06% to 8.14%), and 2.2% for luminal B (from 4.04% to 4.13%).

When assuming that 8.2% of patients received a 5-fraction radiotherapy regimen⁽¹⁴⁾—costlier than the standard 15-fraction course—during the COVID-19 period, and that this practice persisted over five years, the model projected an increase of €1.60 million (to €4.79 million) in incremental breast cancer costs across all subtypes. This translated into a 2.5-fold increase for luminal A, 1.7-fold for luminal B, 1.3-fold for HER2+, and 1.2-fold for TNBC.

Insert Table 7 here

3.5 | Probabilistic Sensitivity Analysis (PSA)

Scatterplots of the 1,000 PSA iterations were generated for each subtype to illustrate the distribution of incremental QALYs and healthcare costs (Figures 4-7). Table 8 presents the mean outcomes from the PSA

by molecular subtype, including incremental QALYs and healthcare costs, as well as the corresponding 95% CIs.

Across all molecular subtypes, the PSA revealed wide uncertainty in incremental QALYs and costs. For luminal A, the 95% CI for incremental QALYs ranged from -1,255.36 to 1,249.99, with a 51% probability of QALY loss, while incremental healthcare costs ranged from -€26,790,493 to €24,962,257, with a 52% probability of cost increase (Figure 4). For luminal B, QALY changes ranged from -625.65 to 604.00 (49% probability of loss) and costs from -€19,523,679 to €19,107,800 (50% probability of increase) (Figure 5). For HER2+, QALYs ranged from -160.11 to 154.10 (50% probability of loss) and costs from -€16,775,684 to €16,469,399 (52% probability of increase) (Figure 6). For TNBC, QALYs ranged from -213.03 to 200.42 (54% probability of loss) and costs from -€47,871,357 to €53,229,055 (51% probability of increase) (Figure 7). These ranges illustrate that, while point estimates align with base-case findings (see Table 6), the CIs encompass both potential gains and losses in QALYs as well as cost savings and increases.

Table 8. Probabilistic sensitivity analysis: QALY and cost changes by breast cancer subtype under care disruption.

Molecular Subtype	Breast cancer cohort	Non-disrupted care (2017-2019)		Disrupted care (2020)		Incremental costs & QALYs (Disrupted care vs. non-disrupted care)	
		QALYs	Cost (€)	QALYs	Cost (€)	ΔQALYs	ΔCost (€)
Luminal A	Total	20 991.0	179 335 236	21 001.1	179 752 285	-15.2	417 049
	95% CI	[20 962.2 ; 21 019.8]	[178 767 318 ; 179,903 154]	[20 972.8 ; 21 029.3]	[179 187 369 ; 180 317 201]	[-1 255.4; 1 249.99]	[-26 790 493 ; 24 962 257]
Luminal B	Total	10 375.6	103 543 022	10 373.5	103 641 533	-2.1	98 511
	95% CI	[10 361.8 ; 10 389.4]	[103 113 677 ; 103 972 368]	[10 360.0 ; 10 386.9]	[103 219 807 ; 104 063 259]	[-625.7 ; 604.0]	[-19 523 679 ; 19 107 800]
HER2+	Total	2 243.4	82 244 443	2 242.3	82 474 170	-1.0	229 727
	95% CI	[2 240.2 ; 2 246.6]	[81 876 267 ; 82 612 619]	[2 238.9 ; 2 245.7]	[82 126 409 ; 82 812 930]	[-160.1 ; 154.1]	[-16 775 684 ; 16 469 399]
Triple-Negative	Total	4 768.7	319 960 341	4 757.1	321 551 351	-11.7	1 591 010
	95% CI	[4 762.5 ; 4 775.0]	[318 901 500 ; 321 019 181]	[4 754.1 ; 4 760.0]	[320 410 320 ; 322 692 381]	[-213.0 ; 200.4]	[-47 871 357 ; 53 229 055]

HER2+: Human Epidermal Growth Factor Receptor 2-positive; QALY: Quality-Adjusted Life Year; 95% CI: 95% confidence interval

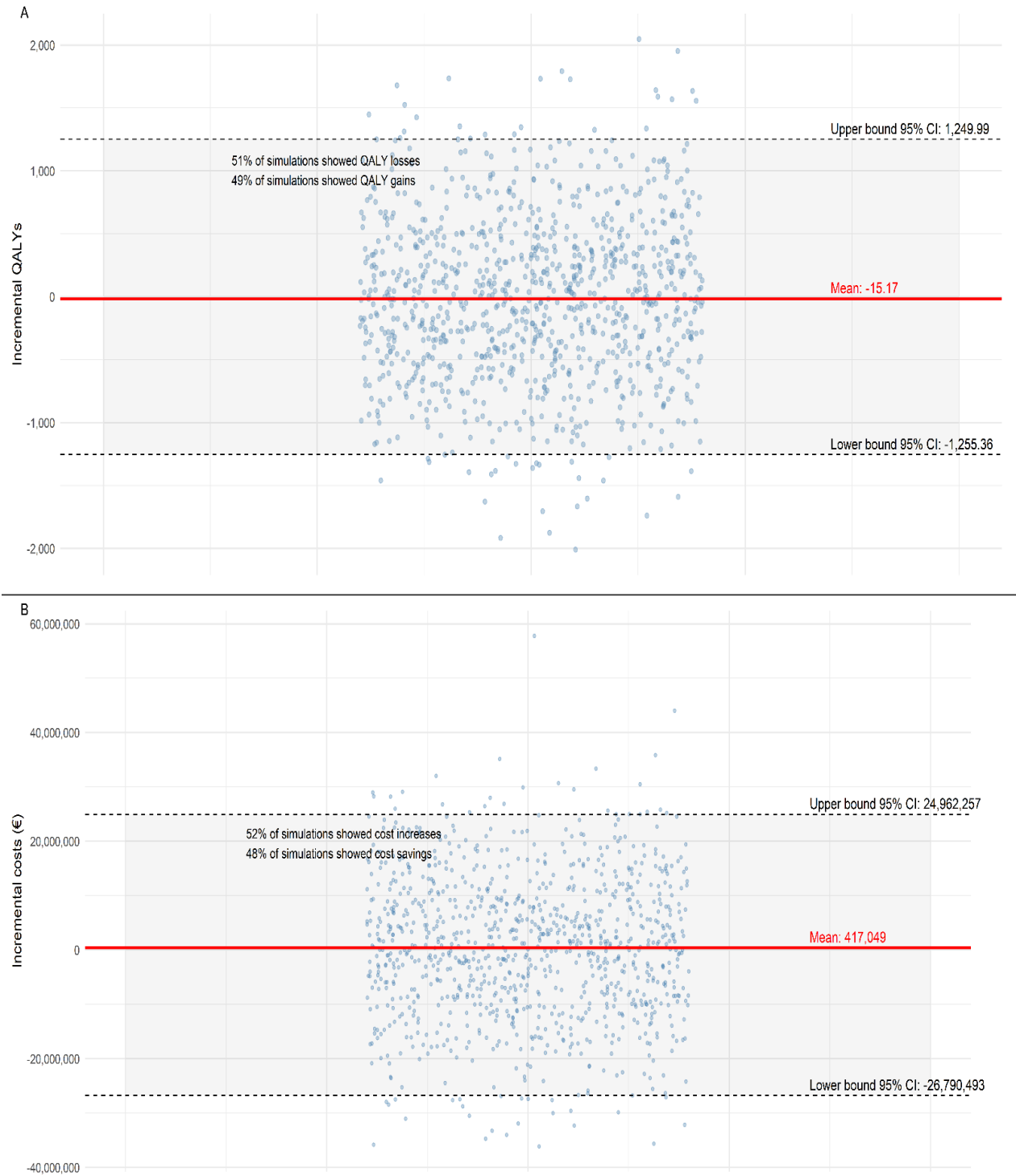


Figure 4. Incremental QALYs (A) and healthcare costs (B) for the total luminal A cohort: Results from 1,000 PSA iterations.

QALY: Quality-Adjusted Life Year; 95% CI: 95% confidence interval

Note. The red solid line represents the mean value across simulations. Black dashed lines indicate the 95% CIs. The shaded grey area marks the CI range. Percentages refer to the proportion of simulations showing gains vs. losses (A) or cost increases vs. savings (B).

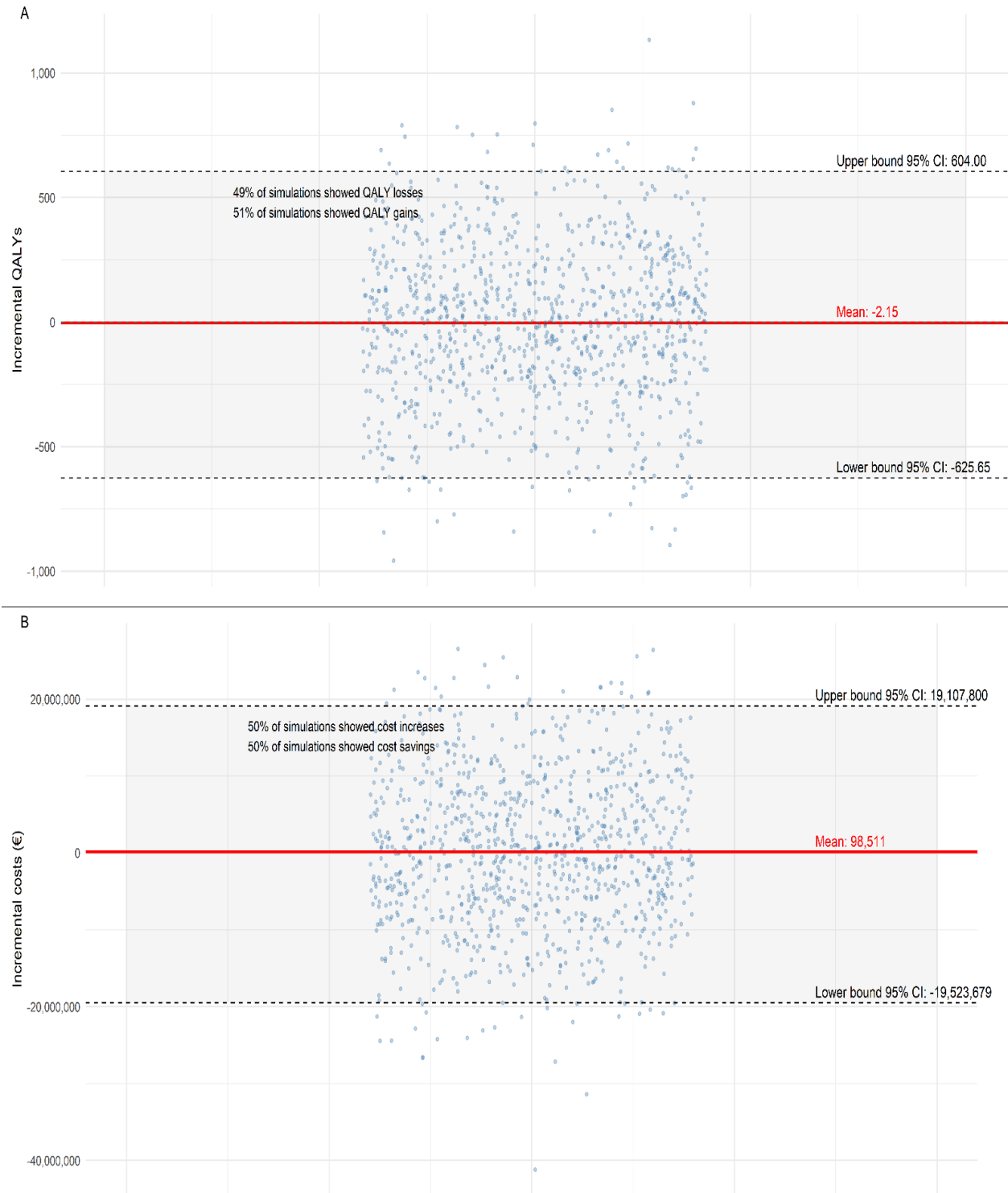


Figure 5. Incremental QALYs (A) and healthcare costs (B) for the total luminal B cohort: Results from 1,000 PSA iterations.

QALY: Quality-Adjusted Life Year; 95% CI: 95% confidence interval

Note. The red solid line represents the mean value across simulations. Black dashed lines indicate the 95% CIs. The shaded grey area marks the CI range. Percentages refer to the proportion of simulations showing gains vs. losses (A) or cost increases vs. savings (B).

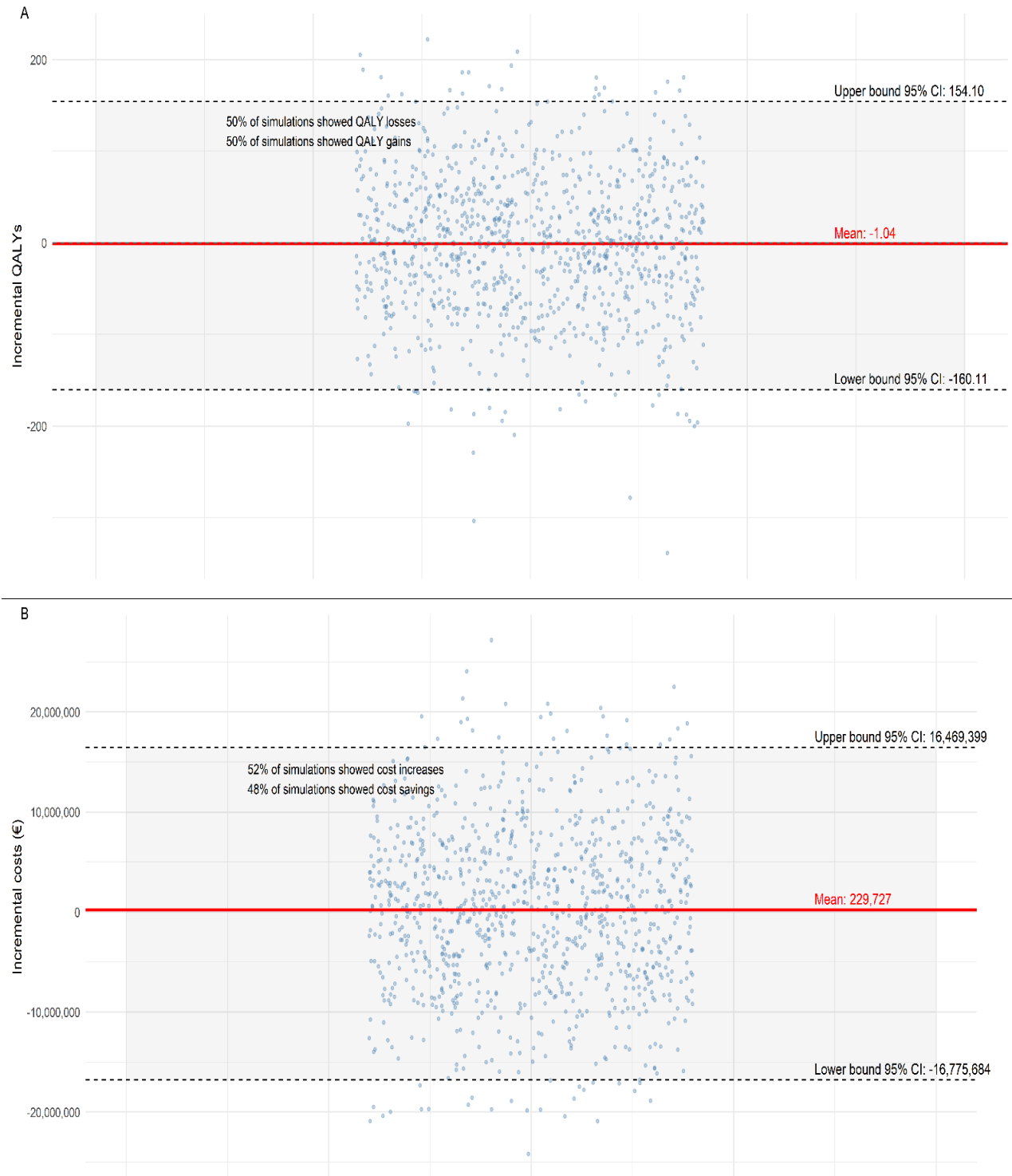


Figure 6. Incremental QALYs (A) and healthcare costs (B) for the total HER2+ cohort: Results from 1,000 PSA iterations.

QALY: Quality-Adjusted Life Year; 95% CI: 95% confidence interval

Note. The red solid line represents the mean value across simulations. Black dashed lines indicate the 95% CIs. The shaded grey area marks the CI range. Percentages refer to the proportion of simulations showing gains vs. losses (A) or cost increases vs. savings (B).

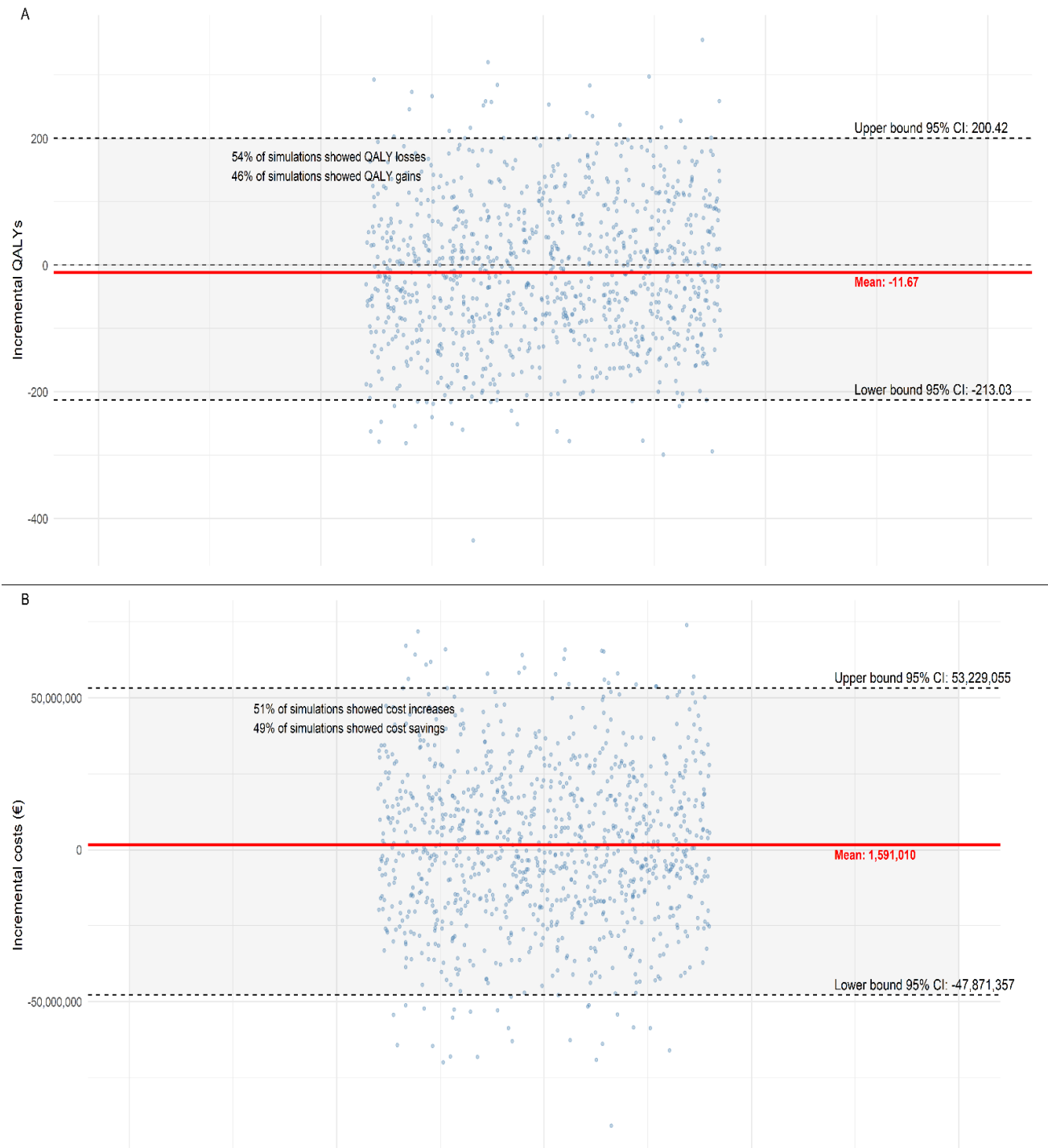


Figure 7. Incremental QALYs (A) and healthcare costs (B) for the total TNBC cohort: Results from 1,000 PSA iterations.

QALY: Quality-Adjusted Life Year; 95% CI: 95% confidence interval

Note. The red solid line represents the mean value across simulations. Black dashed lines indicate the 95% CIs. The shaded grey area marks the CI range. Percentages refer to the proportion of simulations showing gains vs. losses (A) or cost increases vs. savings (B).

4 | Discussion

This modelling exercise projected that breast cancer diagnostic delays between March and June 2020 in Belgium had only a modest health and economic impact over five years. Among 10,147 patients, the 135 undiagnosed cases resulted in an estimated 21 QALYs lost and six additional deaths, with wide uncertainty intervals including zero around the deterministic estimate. This averages to 0.002 QALYs per patient—representing less than one day in full health and falling far below the MCID(59). Additionally, the model projected €3.2 million in extra healthcare costs, representing 1% of Belgian annual breast cancer expenditure. From a public health perspective, the disruption's impact was limited and smaller than expected, despite early reports and widespread concern(60–62) during the first pandemic months.

Diagnostic delays affect all breast cancers, but their impact depends on tumour biology: missing an aggressive subtype such as TNBC or HER2+ carries greater risk than delaying diagnosis of luminal-like cancers(25). In line with this rationale, our model applied clinically supported but hypothesised progression assumptions, which projected heterogeneity by subtype: TNBC and HER2+ together accounted for nearly 70% of excess healthcare costs and 60% of QALY losses.

Extending the time horizon to 10 years yielded similarly limited impacts, with over 32 QALYs lost—equivalent to around one day of full health per patient—and incremental healthcare costs representing <2% (€5.7 million) of Belgian annual breast cancer expenditure. The relatively larger incremental QALY and cost impacts for luminal-like subtypes reflect their longer recurrence trajectories(27,56), whereas most adverse outcomes for HER2+ and TNBC, which recur earlier(27,56), were already captured within the five-year horizon. Even when assuming higher probabilities of UDC being diagnosed at advanced stages across all subtypes, the projected 5-year impact remained modest,

with 37 QALYs lost (under two days in full health per patient) and €9 million in incremental costs (<3% of annual breast cancer expenditure).

Although the deterministic base-case analysis projected modest incremental effects, the PSA revealed that these estimates are highly uncertain. Across all molecular subtypes, the 95% CIs spanned both positive and negative values, with probabilities of QALY loss or cost increase close to 50%. This indicates that the base-case results represent only one possible outcome within a wide distribution: small changes in key parameters could plausibly result in either greater harm or no adverse effect at all. Much of this uncertainty stems from transition probabilities from UDC to stage-specific diagnosis and from utility estimates, particularly for TNBC and HER2+ where data were limited. Consistent with this, the OSA identified TNBC inputs—with especially wide 95% CI—as having the greatest influence on overall results. The robustness of conclusions therefore hinges on the precision of such parameters, underscoring the need for higher-quality, subtype-specific data to reduce uncertainty.

The limited incremental QALYs and healthcare costs, despite wide uncertainty, are likely explained by the small proportion of UDC (1.33%) reported at the end of 2020, which were then modelled over a five-year period. Indeed, even though a larger proportion of patients were undiagnosed during the first few months of the pandemic(14), the majority was recovered by year-end. This is linked to Belgium's screening programmes that resumed in June 2020 and 'do-not-delay-care' campaigns(63), proactive primary-care outreach(64) and teleconsultation policies(65), which supported timely symptomatic presentation. Consistent with this, a Belgian population-level observational study (14) found no evidence of a national stage shift, tumour size increase, or nodal progression in 2020, with DCIS diagnoses—likely most of the missing cases—rebounding by late 2020 and screening volumes recovering in 2021.

Our study is not the first to examine the potential health and economic consequences of COVID-19-related diagnostic delays in breast cancer. Sud et al. (66) modelled delays in the UK's

2-week-wait referral pathway applying per-day hazard ratios for treatment delay uniformly to all stage I–III cases diagnosed through this route, which projected 733 excess breast cancer deaths and 15,339 life-years lost annually in England from a three-month delay. Maringe et al.,(20) using linked NHS registry and hospital data in a stage-shift model, estimated 281–344 additional breast cancer deaths over five years in England due to pandemic-related diagnostic delays. Gheorghe et al.(24) then translated these mortality projections into 4,100 QALYs lost and £23.2 million in productivity losses. Degeling et al.(67), applying an Australian stage-shift model, projected 20–64 life-years lost and AU\$2.0–7.7 million in extra costs for breast cancer under 3–6 month delay scenarios.

Differences between our findings and those of previous modelling studies can be explained by several factors. First, population sizes and healthcare systems differ between Belgium, the UK, and Australia, likely influencing the scale of the impact. Belgium’s smaller population, dense hospital network, and high acute bed capacity(68) supported resilience during the first wave, while rapid measures—such as reimbursed teleconsultations(69), proactive primary care outreach, adapted cancer guidelines(70), and hypofractionated radiotherapy(14)—helped sustain continuity of diagnosis and treatment. Importantly, breast cancer screening resumed in June 2020(14), earlier than in parts of the UK(71,72), which reduced the period of diagnostic delay. Coupled with strong inter-hospital collaboration(73) and national ‘do-not-delay-care’ campaigns, these factors limited the proportion of undiagnosed cases that progressed to later stages. This helps explain why our model projected smaller QALY losses, lower additional healthcare costs, and fewer excess deaths compared with modelling studies from the UK and Australia. In those regions, screening interruptions persisted longer and system capacity was more constrained.

Second, prior models varied substantially in design: Sud et al.(66) applied per-day hazards of delay across all stage I–III cancers diagnosed via the 2-week-wait pathway; Maringe et al.(20) modelled large shifts from routine or screening routes to emergency presentations; Gheorghe et al.(24) translated Maringe et al.(20) excess deaths into QALY and productivity losses using a societal human-capital approach; and

Degeling et al.(67) applied an inverse stage-shift method, conservatively modelling stage I to II progression under uniform 3-6 month delays. By contrast, our subtype-specific Markov model compared two Belgian cohorts: the observed 2020 breast cancer population (with 1.33% UDC) and a projected 'no disruption' cohort based on 2017–2019 trends, with UDC progression under diagnostic delay hypothesised to stage-specific diagnoses according to molecular subtype. Third, the duration and intensity of disruption differed markedly: Belgian screening was suspended for approximately three months with rapid recovery, whereas UK scenarios assumed screening disruptions lasting up to 12 months(66). Finally, our analysis adopted a healthcare payer perspective, excluded DCIS and unknown-stage cases, and assumed constant treatment trajectories. These choices, combined with Belgium's mitigation measures and the relatively rapid resumption of screening(14)—in contrast to Scotland(72) and Wales(71) for instance, where screening remained suspended until August 2020—may help explain the smaller health and economic impacts observed.

While this study focuses on the incremental healthcare costs (€3.2 million) and QALY losses (21 QALYs) associated with delayed breast cancer diagnosis during the first pandemic wave, it is important to recognise that the screening suspension also likely generated short-term reductions in resource use and patient burden. Some of the financial reduction was a deferral of expenditure, as the catch-up phase later in 2020 reinstated most screening costs(14). However, genuine savings and avoided QALY losses likely arose from the temporary decline in false-positive work-ups and the avoidance of overdiagnosis of indolent tumours(74).

From a societal perspective, short-term economic effects may have included fewer lost workdays associated with screening attendance and follow-up visits, although potential longer-term productivity losses due to delayed diagnoses cannot be excluded. These effects could not be quantified, as no data exist on how temporary screening pauses affect absenteeism or anxiety at the population level. The implications for anxiety are more complex: while women may have avoided distress associated with

false-positive recalls (75), this may have been offset by short-term pandemic-related stress and uncertainty surrounding postponed care.

While these aspects were not included in our healthcare payer model, they indicate that the broader societal impact of the screening disruption may be more nuanced than direct medical costs alone suggest, and must be weighed against the burden of additional advanced-stage aggressive cancers.

4.1 | Strengths and limitations

The primary strength of this study lies in its use of a Markov model that integrates Belgian observational data on breast cancer incidence by clinical stage. To our knowledge, this is the first subtype-specific Markov model assessing the health and economic impact of COVID-19–related breast cancer diagnostic delays in Belgium. An additional strength is the explicit stratification by molecular subtype, a level of clinical detail rarely included in international COVID-19 modelling studies(20,24,66,67,76–78). By accounting for tumour heterogeneity rather than treating breast cancer as a single entity, the model can reveal differential impacts across subtypes(12,25)—particularly for more aggressive forms such as TNBC, which are highly sensitive to even short diagnostic delays. Despite uncertainties in input parameters, robust sensitivity analyses and expert consultations strengthened the model’s credibility and ensured that limitations were explicitly addressed. The framework is adaptable to other countries, providing a versatile policy tool (28). Several limitations should be noted. While stage-specific projections from the BCR enabled estimation of UDC by stage in 2020, data were not subtype-specific. Applying the overall stage distribution uniformly across subtypes may overestimate the contribution of aggressive cancers, which are less likely to remain undiagnosed. Conversely, it may underestimate their impact, as missed cases would more plausibly present at later stages. A related limitation is that the model assumed transition probabilities from UDC to stage-specific diagnosis across molecular subtypes. The registry data did not allow us to distinguish diagnoses arising from resumed screening versus symptomatic

presentations. National data, however, indicate that the recovery in late 2020 was driven by early-stage tumours in the screening age group(14), suggesting that screening activity primarily accounted for the catch-up. We therefore hypothesised that luminal-like subtypes would be diagnosed mostly at early stages, while aggressive subtypes would be more likely to present at advanced stages, reflecting symptomatic interval cancers. Some input parameters—such as transition probabilities from UDC to stage-specific diagnosis by subtype and selected disutilities (Table 4)—were not empirically available and were instead specified as plausible assumptions informed by clinical evidence. While this approach enabled model implementation, it inevitably introduces additional uncertainty in the incremental outcomes, as reflected in the PSA. Simplifying treatment trajectories (e.g., equating costs and utilities between luminal subtypes, not accounting for the modest increase in NAT use for early-stage disease(14)) may have attenuated subgroup-specific differences, producing more moderate aggregate results than would be expected if full clinical heterogeneity were incorporated. While age is a key prognostic factor and BCR data are available by age group, the absence of stratification by subtype prevented the model from accounting for age-related differences in progression, treatment response, or survival within subtypes—for instance, potentially larger QALY and healthcare cost impacts among younger women with aggressive cancers, and smaller impacts among older women due to higher competing mortality and less intensive treatment. The model assumed identical input parameters for disrupted and non-disrupted care, except for UDC and related transition probabilities. A third scenario incorporating Belgian data on radiotherapy reimbursement(79) (5-fraction vs. 15-fraction) showed that altering a single parameter could change incremental costs. Other adaptations—telemedicine(65), early discharge(80), and organizational changes—also likely affected costs and outcomes in both directions, so cumulative unmodelled effects may have differed from our estimates. Finally, certain clinical aspects were not modelled to avoid added complexity—for example, recurrence risks vary by stage and subtype, but LRR within stages I–III was not modelled separately.

4.2 | Policy implications and future research

Although concerns during the initial pandemic suggested severe consequences, our analysis and BCR data indicate that the early disruption of breast cancer diagnosis had a modest five-year impact that, while growing slightly when projected over a 10-year horizon, remained limited overall. This likely reflects the rapid recovery of screening, proactive GP outreach, and targeted “do-not-delay-care” campaigns, which helped avoid a sustained stage shift. For policy, these findings highlight the need to protect continuity of diagnostic services during system shocks, minimise backlogs once services resume, and ensure vigilance for aggressive tumours such as TNBC and HER2+.

The modest aggregate impact observed following the short-term suspension of breast cancer screening should be interpreted with caution and does not, in itself, provide evidence to support changes to existing screening policies. While these findings primarily reflect the resilience of the healthcare system and timely recovery measures, they are also consistent with hypotheses proposed in the literature suggesting that brief interruptions may preferentially reduce the detection of indolent tumours, with limited short-term consequences for aggressive disease. Importantly, this interpretation is exploratory and hypothesis-generating, given the short duration of the disruption and the substantial uncertainty surrounding the estimates. Any consideration of optimising screening strategies, including approaches that account for tumour aggressiveness, would therefore require stronger empirical evidence from dedicated studies and improved data infrastructure. As noted in the limitations, the BCR does not routinely report incidence by molecular subtype, necessitating estimation in this study; systematic capture of subtype-specific incidence would be essential for future evaluations of screening efficiency.

Multi-state modelling is increasingly used in health-economic research because it can represent disease progression and time-dependent transitions more realistically than conventional cohort models (e.g., partitioned survival models, 3-state Markov models). In this study, it allowed us to explicitly model

transitions from an ‘undiagnosed’ state to specific clinical stages and to quantify the downstream effects of diagnostic delay on recurrence and mortality in a detailed way. This flexible structure also supported scenario analyses—such as extending the time horizon, increasing transition probabilities from UDC towards advanced-stage disease across molecular subtypes, and modelling shifts in radiotherapy costs during COVID-19—making it a useful framework for evaluating health system resilience.

Future work should build on these insights by maintaining registry-based monitoring to capture possible longer-term impacts, particularly in aggressive subtypes, and by incorporating outcomes such as recurrence, overall survival, and HRQoL. An important next step will be for cancer registries to routinely provide subtype-specific incidence and stage data, enabling more granular assessment of differential impacts and supporting robust, policy-relevant modelling. It will also be important to evaluate the cumulative impact of multiple pandemic waves and to assess the effects of treatment adaptations (e.g., telemedicine, altered radiotherapy regimens, early discharge) that may have influenced QALY and healthcare cost changes. Finally, this study illustrates the value of rapid modelling exercises as a complement to slower-to-emerge observational data, providing policymakers with timely evidence to guide mitigation strategies and strengthen preparedness for future health system disruptions.

4.3 | Conclusion

This study shows that breast cancer diagnostic delays during Belgium’s first COVID-19 wave were associated with modest five-year health and economic consequences, estimated at 21 QALYs lost, six additional deaths, and €3.2 million in additional healthcare costs (<1% of annual expenditure). These smaller-than-expected impacts are reassuring and likely reflect Belgium’s rapid recovery of screening services and mitigation measures, which limited sustained stage shifts. Subtype-specific modelling nevertheless indicated that aggressive cancers, such as TNBC and HER2+, are more sensitive to diagnostic delays than luminal-like subtypes. However, projections were strongly assumption-driven and

characterised by substantial uncertainty, underscoring the need for higher-quality, subtype-specific data. Continued monitoring of recurrence, survival, and HRQoL is therefore essential to capture potential longer-term effects. Ultimately, generating robust subtype-specific evidence will be crucial to inform future research on whether more tailored screening and diagnostic strategies could improve resilience during systemic healthcare disruptions.

LIST OF ABBREVIATIONS

AT: Adjuvant Therapy

BCPI: Belgian Centre for Pharmacotherapeutic Information

BCR: Belgian Cancer Registry

BCS: Breast-Conserving Surgery

CDK4/6: Cyclin-Dependent Kinases 4 and 6

COVID-19: Coronavirus Disease 2019

D&T: Diagnosis and Treatment

DCIS: Ductal Carcinoma In Situ

ER: Estrogen Receptor

ESMO: European Society for Medical Oncology

FU: Follow-Up

HER2: Human Epidermal Growth Factor Receptor 2

HRQoL: Health-Related Quality of Life

LRR: Locoregional Recurrence

MCID: Minimal Clinically Important Difference

MST: Mastectomy

NIHDI: National Institute for Health and Disability Insurance

OSA: One-Way Sensitivity Analysis

PR: Progesterone Receptor

PSA: Probabilistic Sensitivity Analysis

QALY: Quality-Adjusted Life Year

RT: Radiotherapy

SE: Standard Error

TNBC: Triple-Negative Breast Cancer

UDC: Undiagnosed Cancer

VBA: Visual Basic for Applications

DECLARATIONS

Ethics approval and consent to participate: This study was based on aggregated, de-identified registry data from the Belgian Cancer Registry (BCR), which exempted it from ethical review.

Consent for publication: Not applicable.

Availability of data and materials: The Excel-based Markov model developed and analysed during the current study is provided as supplementary material with this article. In addition, a Word technical appendix (Additional file 1) is provided, containing detailed descriptions of the model structure, transition probabilities, utilities, healthcare costs, and assumptions used in the analysis.

Competing interests: The authors declare that they have no competing interests.

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Authors' contributions: YK conceived and designed the study, developed the Markov model, collected data inputs, performed the analyses, and drafted the manuscript. NV supervised the economic modelling, and critically reviewed the manuscript. CM contributed to the study design, provided clinical expertise, and validated treatment pathways. KV provided input on epidemiological aspects and critically reviewed the manuscript. SG critically reviewed the manuscript. BD critically reviewed the manuscript. FV and HP ensured oncological accuracy in the model and critically reviewed the manuscript. DS supervised the project, provided methodological guidance, and critically reviewed the manuscript.

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Table 4. Baseline input parameters for non-disrupted care (pre-COVID-19) and disrupted care (COVID-19) cohorts by molecular subtype

Input parameters	Subtype	<i>Transition probabilities</i>				Source(s)
		Base Value	Standard Error	OSA – 95% CI Lower Bound	OSA – 95% CI Upper Bound	
Stage I D & T to Dead	Luminal A	0.002	0.001	0.001	0.004	(81)
	Luminal B	0.010	0.002	0.006	0.013	
	HER2+	0.016	0.002	0.012	0.021	
	TNBC	0.017	0.002	0.012	0.02	
Stage II D & T to Dead	Luminal A	0.005	0.001	0.002	0.007	
	Luminal B	0.012	0.002	0.008	0.016	
	HER2+	0.019	0.003	0.014	0.024	
	TNBC	0.019	0.003	0.014	0.024	
Stage III D & T to Dead	Luminal A	0.012	0.002	0.008	0.016	
	Luminal B	0.020	0.003	0.015	0.025	
	HER2+	0.026	0.003	0.020	0.032	
	TNBC	0.027	0.003	0.021	0.032	
Stage I-III FU to Stage I-III LRR	Luminal A	0.003	0.001	0.001	0.004	(27)
	Luminal B	0.005	0.002	0.002	0.008	
	HER2+	0.006	0.003	0.000	0.013	
	TNBC	0.007	0.003	0.002	0.012	
Stage I-III FU to Stage IV D & 1L	Luminal A	0.005	0.001	0.003	0.007	
	Luminal B	0.011	0.002	0.007	0.016	
	HER2+	0.015	0.005	0.004	0.025	
	TNBC	0.013	0.004	0.006	0.020	
Stage I-III FU to Dead	Luminal A	0.001	0.000	0.000	0.001	
	Luminal B	0.001	0.001	0.000	0.002	
	HER2+	0.001	0.002	0.000	0.006	
	TNBC	0.002	0.001	0.000	0.004	
Stage I-III LRR to Dead	Luminal A	0.011	0.002	0.008	0.014	(82)
	Luminal B	0.032	0.005	0.023	0.042	

	HER2+	0.020	0.004	0.013	0.028	
	TNBC	0.035	0.006	0.023	0.046	
Stage IV D & 1L to Stage IV 2L	Luminal A	0.060	0.005	0.051	0.069	(83)
	Luminal B	0.060	0.005	0.051	0.069	
	HER2+	0.054	0.008	0.039	0.069	
	TNBC	0.062	0.012	0.038	0.085	
Stage IV D & 1L to Dead	Luminal A	0.005	0.001	0.003	0.008	
	Luminal B	0.005	0.001	0.003	0.008	
	HER2+	0.007	0.004	0.000	0.014	
	TNBC	0.012	0.005	0.002	0.023	
Stage IV 2L to Stage IV 3L	Luminal A	0.073	0.006	0.061	0.085	
	Luminal B	0.073	0.006	0.061	0.085	
	HER2+	0.065	0.010	0.045	0.086	
	TNBC	0.065	0.014	0.037	0.093	
Stage IV 2L to Dead	Luminal A	0.008	0.002	0.004	0.012	
	Luminal B	0.008	0.002	0.004	0.012	
	HER2+	0.007	0.004	0.000	0.014	
	TNBC	0.013	0.007	0.000	0.026	
Stage IV 3L to Stage IV 4L	Luminal A	0.073	0.007	0.059	0.087	
	Luminal B	0.073	0.007	0.059	0.087	
	HER2+	0.063	0.013	0.039	0.088	
	TNBC	0.057	0.016	0.026	0.087	
Stage IV 3L to Dead	Luminal A	0.009	0.003	0.004	0.014	
	Luminal B	0.009	0.003	0.004	0.014	
	HER2+	0.009	0.005	0.000	0.018	
	TNBC	0.018	0.009	0.000	0.035	
Stage IV 4L to Dead	Luminal A	0.010	0.003	0.004	0.015	
	Luminal B	0.010	0.003	0.004	0.015	
	HER2+	0.010	0.010	0.000	0.031	
	TNBC	0.022	0.010	0.003	0.041	
UDC to Stage I D & T	Luminal A	<u>0.400</u>	0.020	0.361	0.439	Hypothesize d based on clinical evidence of stage
	Luminal B	<u>0.300</u>	0.020	0.261	0.389	
	HER2+	<u>0.100</u>	0.020	0.061	0.139	

	TNBC	<u>0.050</u>	0.020	0.011	0.089	progression after diagnostic delay (25)		
UDC to Stage II D & T	Luminal A	<u>0.400</u>	0.020	0.361	0.439			
	Luminal B	<u>0.350</u>	0.020	0.311	0.389			
	HER2+	<u>0.150</u>	0.020	0.111	0.189			
	TNBC	<u>0.050</u>	0.020	0.011	0.089			
UDC to Stage III D & T	Luminal A	<u>0.100</u>	0.020	0.061	0.139			
	Luminal B	<u>0.150</u>	0.020	0.111	0.189			
	HER2+	<u>0.300</u>	0.020	0.261	0.339			
	TNBC	<u>0.100</u>	0.020	0.061	0.139			
UDC to Stage IV D & 1L	Luminal A	<u>0.080</u>	0.020	0.041	0.119			
	Luminal B	<u>0.150</u>	0.020	0.111	0.189			
	HER2+	<u>0.350</u>	0.020	0.311	0.389			
	TNBC	<u>0.750</u>	0.020	0.711	0.789			
UDC to Dead	Luminal A	<u>0.010</u>	0.020	0.000	0.049			
	Luminal B	<u>0.025</u>	0.020	0.000	0.064			
	HER2+	<u>0.035</u>	0.020	0.000	0.074			
	TNBC	<u>0.040</u>	0.020	0.001	0.079			
Utilities								
Input parameters	Subtype	Base Value	Standard Error	Disutility value	OSA – 95% CI Lower Bound	OSA – 95% CI Upper Bound	Source(s)	
UDC	Luminal A	0.82	0.005	-	0.811	0.829	(36)	
	Luminal B	0.82	0.005		0.811	0.829		
	HER2+	0.82	0.005		0.811	0.829		
	TNBC	0.82	0.005		0.811	0.829		
Stage I D & T	Luminal A	0.787	0.022	0.033	0.744	0.829	(37)	
	Luminal B	0.787	0.022		0.744	0.829		
	HER2+	<u>0.770</u>	0.077		0.620	0.920		(41–43)
	TNBC	<u>0.755</u>	0.077		0.605	0.905		(43,44)
Stage II D & T	Luminal A	0.787	0.022	0.033	0.744	0.829	(37)	
	Luminal B	0.787	0.022		0.744	0.829		
	HER2+	<u>0.770</u>	0.077		0.620	0.920		(41–43)
	TNBC	<u>0.755</u>	0.077		0.605	0.905		(43,44)
Stage III D & T	Luminal A	0.782	0.025	0.038	0.733	0.831	(37)	
	Luminal B	0.782	0.025		0.733	0.831		

	HER2+	<u>0.760</u>	0.077	0.060	0.610	0.910	(41–43)
	TNBC	<u>0.745</u>	0.077	0.075	0.95	0.895	(43,44)
Stage I-III FU (adjuvant therapy)	Luminal A	0.772	0.021	0.048	0.731	0.812	(37)
	Luminal B	0.772	0.021	0.048	0.731	0.812	
	HER2+	0.733	0.041	0.087	0.654	0.813	
	TNBC	0.733	0.041	0.087	0.654	0.813	
Stage I-III FU (post-adjuv ant therapy)	Luminal A	0.814	0.005	0.006	0.805	0.823	(37)
	Luminal B	0.814	0.005	0.006	0.805	0.823	
	HER2+	0.814	0.005	0.006	0.805	0.823	
	TNBC	0.814	0.005	0.006	0.805	0.823	
Stage I-III LRR	Luminal A	0.770	0.007	0.050	0.757	0.783	(38)
	Luminal B	0.770	0.007	0.050	0.757	0.783	
	HER2+	<u>0.730</u>	0.077	0.090	0.580	0.880	
	TNBC	<u>0.710</u>	0.077	0.110	0.560	0.860	
Stage IV D & T	Luminal A	0.730	0.009	0.090	0.712	0.748	(38)
	Luminal B	0.730	0.009	0.090	0.712	0.748	
	HER2+	<u>0.690</u>	0.077	0.130	0.540	0.840	(41)
	TNBC	<u>0.670</u>	0.077	0.150	0.520	0.820	(47)
Stage IV 2L	Luminal A	<u>0.720</u>	0.077	0.100	0.570	0.870	(48)
	Luminal B	<u>0.720</u>	0.016	0.100	0.570	0.870	
	HER2+	<u>0.680</u>	0.077	0.140	0.530	0.830	(49)
	TNBC	<u>0.650</u>	0.077	0.170	0.500	0.800	(50)
Stage IV 3L	Luminal A	<u>0.700</u>	0.077	0.120	0.550	0.850	(49)
	Luminal B	<u>0.700</u>	0.077	0.120	0.550	0.850	
	HER2+	<u>0.680</u>	0.077	0.140	0.530	0.830	(51)
	TNBC	<u>0.630</u>	0.077	0.190	0.480	0.780	(52)
Stage IV 4L	Luminal A	<u>0.690</u>	0.077	0.130	0.540	0.840	(53,84)
	Luminal B	0.690	0.077	0.130	0.540	0.840	
	HER2+	<u>0.660</u>	0.077	0.160	0.510	0.810	(54)
	TNBC	<u>0.600</u>	0.077	0.220	0.450	0.750	(55)
<i>Direct healthcare costs</i>							
Input parameters	Subtype	Base Value	Standard Error	OSA – 95% CI Lower Bound		OSA – 95% CI Upper Bound	Source(s)
Stage I D & T	Luminal A	4 548	455	3 657		5 440	NomenSoft nomenclatu re codes
	Luminal B	4 548	455	3 657		5 440	

	HER2+	7 563	756	6 081	9 046	(see the Additional file 1)
	TNBC	28 352	2 835	22 795	33 910	
Stage II D & T	Luminal A	4 548	455	3 657	5 440	
	Luminal B	4 548	455	3 657	5 440	
	HER2+	12 601	1 260	10 131	15 071	
	TNBC	28 352	2 835	22 795	33 910	
Stage III D & T	Luminal A	5 523	552	4 441	6 606	
	Luminal B	5 523	552	4 441	6 606	
	HER2+	13 576	1 358	10 915	16 236	
	TNBC	29 327	2 933	23 579	35 075	
Stage I-III FU (adjuvant therapy)	Luminal A	536	54	431	641	BCPI(34)
	Luminal B	536	54	431	641	
	HER2+	16 105	1 610	12 948	19 262	
	TNBC	23 804	2 380	19 139	28 470	
Stage I-III FU (post-adjuv ant therapy)	Luminal A	334	33	269	400	NomenSoft nomenclatu re codes (see the Additional file 1)
	Luminal B	334	33	269	400	
	HER2+	334	33	269	400	
	TNBC	334	33	269	400	
Stage I-III LRR	Luminal A	4 873	487	3 918	5 828	Mean Stage I-III D&T costs
	Luminal B	4 873	487	3 918	5 828	
	HER2+	11 247	1 125	9 042	13 451	
	TNBC	28 677	2,868	23 057	34 298	
Stage IV D & T	Luminal A	5 934	593	4 771	7 098	BCPI(34)
	Luminal B	5 934	593	4 771	7 098	
	HER2+	16 385	1 638	13 173	19 596	
	TNBC	24 649	2 465	19 818	29 480	
Stage IV 2L	Luminal A	5 922	592	4 762	7 083	
	Luminal B	5 922	592	4 762	7 083	
	HER2+	14 537	1 454	11 688	17 386	
	TNBC	25 038	2 504	20 131	29 945	
Stage IV 3L	Luminal A	28 976	2 898	23 297	34 655	
	Luminal B	28 976	2 898	23 297	34 655	
	HER2+	4 768	477	3 833	5 702	
	TNBC	9 374	937	7 537	11 211	
Stage IV 4L	Luminal A	1 261	126	1 014	1 509	

	Luminal B	1 261	126	1 014	1 509		
	HER2+	19 663	1 966	15 809	23 518		
	TNBC	1 245	124	1 001	1 488		
Scenario Analysis							
Input parameters	Subtype	Base Value	Standard Error	Scenario Analysis - Multiplier	OSA – 95% CI Lower Bound	OSA – 95% CI Upper Bound	Source(s)
UDC to Stage I D & T	Luminal A	<u>0.10</u>	0.001	0.25	0.098	0.102	Hypothesized based on clinical evidence of stage progression after diagnostic delay (25)
	Luminal B	<u>0.09</u>	0.001	0.30	0.088	0.092	
	HER2+	<u>0.05</u>	0.001	0.50	0.049	0.051	
	TNBC	<u>0.01</u>	0.000	0.20	0.009	0.011	
UDC to Stage II D & T	Luminal A	<u>0.10</u>	0.001	0.25	0.098	0.102	
	Luminal B	<u>0.12</u>	0.001	0.35	0.120	0.125	
	HER2+	<u>0.05</u>	0.001	0.35	0.051	0.054	
	TNBC	<u>0.01</u>	0.000	0.20	0.009	0.011	
UDC to Stage III D & T	Luminal A	<u>0.25</u>	0.001	2.50	0.247	0.253	
	Luminal B	<u>0.30</u>	0.001	2.00	0.297	0.303	
	HER2+	<u>0.35</u>	0.002	1.15	0.342	0.348	
	TNBC	<u>0.05</u>	0.001	0.50	0.039	0.041	
UDC to Stage IV D & 1L	Luminal A	<u>0.52</u>	0.002	6.50	0.517	0.523	
	Luminal B	<u>0.45</u>	0.002	3.00	0.447	0.453	
	HER2+	<u>0.46</u>	0.002	1.30	0.452	0.458	
	TNBC	<u>0.86</u>	0.001	1.15	0.860	0.865	
UDC to Dead	Luminal A	<u>0.01</u>	0.000	1.00	0.009	0.011	
	Luminal B	<u>0.03</u>	0.000	1.00	0.024	0.026	
	HER2+	<u>0.04</u>	0.001	1.00	0.034	0.036	
	TNBC	<u>0.04</u>	0.001	1.00	0.039	0.041	

D: Diagnosis; FU: Follow-Up; LRR: Locoregional Recurrence; OSA: One-way Sensitivity Analysis; T: Treatment; TNBC: Triple-Negative Breast Cancer; UDC: Undiagnosed Cancer; 1L: 1st systemic therapy line; 2L: 2nd systemic therapy line; 3L: 3rd systemic therapy line; 4L: 4th systemic therapy line

* Bold input parameters represent those only present during the disrupted care situation (COVID-19)

** 95% CI: 95% confidence interval, which was calculated as follows: Base case value $\pm$ 1.96 x SE

*** Standard errors were calculated using $SE = SD/\sqrt{n}$ for means and $SE = \sqrt{p(1 - p)/n}$ for proportions

**** Hypothesized input values were underlined

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Table 6. Base case: Five-year QALY and cost changes by breast cancer subtype under care disruption.

Molecular Subtype	Cohort (n)	Non-disrupted care (2017-2019)			Disrupted-care (2020)			Incremental costs & QALys (Disrupted-care vs. Non-disrupted care)			
		QALys	Cost (€)	Death %	QALys	Cost (€)	Death %	ΔQALys	ΔCost (€)	Δ% Point	Additional deaths
Luminal A	Total cohort (n = 5 479)	20 987.5	179 522 108	2.80	20 982.7	180 004 052	2.88	-4.8	481 944	0.08	2
	Per patient	3.831	32 765		3.830	32 853		-0.001	88		
Luminal B	Total cohort (n= 2 740)	10 378.9	103 245 734	4.04	10 375.0	103 770 423	4.09	-4.0	524 689	0.05	1
	Per patient	3.788	37 681		3.786	37 872		-0.001	191		
HER2+	Total cohort (n = 609)	2 242.4	82 155 748	8.06	2 240.0	82 702 749	8.13	-2.5	547 000	0.07	1
	Per patient	3.682	134 903		3.678	135 801		-0.004	898		
TNBC	Total cohort (n = 1 319)	4 764.4	319 433 158	10.81	4 754.9	321 072 481	11.01	-9.5	1 639 323	0.20	2
	Per patient	3.612	242 178		3.605	243 421		-0.007	1 243		
Breast cancer	Total cohort (n = 10 147)							-20.7	3 192 956		
	Per patient							-0.002	315		

HER2+: Human Epidermal Growth Factor Receptor 2–positive; QALY: Quality-Adjusted Life Year; TNBC: Triple-Negative Breast Cancer

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Table 7. Scenario analysis: Incremental changes in QALYs and healthcare costs across three scenarios, by molecular subtype.

Scenario type	Molecular Subtype	Cohort (n)	Non-disrupted care (2017-2019)			Disrupted care (2020)			Incremental healthcare costs & QALYs (Disrupted care vs. Non-disrupted care)			
			QALYs	Cost (€)	Death %	QALYs	Cost (€)	Death %	ΔQALYs	ΔCost (€)	Δ% Point	Additional deaths

10-year time horizon	Luminal A	Total	39 531.7	342 044 246	6.36	39 517.6	343 230 607	6.40	-13.1	1 186 360	0.04	3	
		Per patient	7.215	62 428		7.213	62 645		-0.002	217			
	Luminal B	Total	19 182.2	228 963 281	9.10	19 172.2	230 020 328	9.16	-10.0	1 057 047	0.06	2	
		Per patient	7.001	83 563		6.997	83 949		-0.004	386			
	HER2+	Total	4 103.1	128 324 067	13.64	4 097.5	129 197 648	13.75	-5.5	873 581	0.11	1	
		Per patient	6.737	210 713		6.728	212 147		-0.009	1,434			
	TNBC	Total	8 538.1	431 548 773	20.44	8 534.6	434 122 775	20.77	-3.4	2 574 002	0.33	4	
		Per patient	6.473	327 179		6.471	329 130		-0.003	1 951			
	Breast Cancer	Total								-32.1	5 690 990		
		Per patient								-0.003	561		

Shift to advance d stages for the UDC patients	Luminal A	Total	20 987.5	179 522 108	2.80	20 972.4	184 012 139	2.88	-15.2	4 490 031	0.08	5
		Per patient	3.831	32 765		3.828	33 585		-0.003	819		
	Luminal B	Total	10 378.9	103 245 734	4.04	10 371.2	105 063 870	4.13	-7.7	1 818 135	0.09	2
		Per patient	3.788	37 681		3.785	38 344		-0.003	664		

	HER2+	Total	2 242.4	82 155 748	8.06	2 239.5	82 838 914	8.14	-2.9	683 165	0.08	1	
		Per patient	3.682	134 903		3.677	136 024		-0.005	1 122			
	TNBC	Total	4 764.4	319 433 158	10.81	4 753.3	321 443 662	11.03	-11.1	2 010 504	0.22	3	
		Per patient	3.612	242 178		3.604	243 703		-0.008	1 524			
	Breast Cancer	Total	ARTICLE IN PRESS							-36.9	9 001 835		
		Per patient								-0.004	887		
Shift in radiotherapy costs during COVID-19	Luminal A	Total	20 987.5	179 522 108	2.80	20 982.7	180 740 027	2.88	-4.8	1 217 920	0.08	2	
		Per patient	3.831	32 765		3.830	32 988		-0.001	222			
	Luminal B	Total	10 378.9	103 245 734	4.04	10 375.0	104 136 793	4.09	-3.9	891 059	0.05	1	
		Per patient	3.788	37 681		3.786	38 006		-0.001	325			
	HER2+	Total	2 242.4	82 155 748	8.06	2 240.0	82 856 543	8.13	-2.5	700 794	0.07	1	
		Per patient	3.682	134 903		3.678	136 053		-0.004	1 151			
	TNBC	Total	4 764.4	319 433 158	10.81	4 754.9	321 413 901	11.01	-9.5	1 980 743	0.20	2	
		Per patient	3.612	242 178		3.605	243 680		-0.007	1 502			

Breast Cancer	Total		-20.7	4 790 516	
	Per patient		-0.002	472	

HER2+: Human Epidermal Growth Factor Receptor 2-positive; QALY: Quality-Adjusted Life Year; TNBC: Triple-Negative Breast Cancer; UDC: Undiagnosed Cancer

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